

Weak Saturation of the 3-Cube in Complete Graphs: Exact Values at Orders 9, 10, 11

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Abstract

Let Q_3 be the three-dimensional cube. The weak saturation number of Q_3 in complete graphs is determined exactly at three consecutive orders:

$$\text{wsat}(K_9, Q_3) = 16, \quad \text{wsat}(K_{10}, Q_3) = 18, \quad \text{wsat}(K_{11}, Q_3) = 19.$$

The value at $n = 9$ gives a counterexample to the formula $\text{wsat}(K_n, Q_3) = 2n - 1$ reported as a conjecture in the 2025 Park City Mathematics Institute project slides of Trakulthongchai, Xu, and Zhu. The $n = 9$ result is established by an exact closure computation: after normalizing the first edge of a saturation order, the problem becomes the search for minimum generating sets in a monotone closure system on the $\binom{9}{2} - 12 = 24$ edges outside a fixed cube. All $\binom{24}{4} = 10,626$ four-edge seeds are excluded, while a five-edge seed percolates. An independent Go implementation uses explicit edge lists and queue-indexed implications and reproduces the lower bound. A separate graph-level verifier enumerates all 7,560 labelled copies of Q_3 in K_9 and checks the complete certificate.

For $n = 10$, all $\binom{33}{6} = 1,107,568$ six-edge seeds are reduced by the stabilizer of the marked cube edge to 146,131 orbit representatives, none of which percolates. For $n = 11$, the seven-edge orbit layer contains 1,498,208 representatives; none percolates. Burnside's lemma independently gives the $n = 11$ orbit counts. Explicit certificates of sizes 16, 18, and 19 provide the matching upper bounds. The certificate tables record, for every edge addition, an injective vertex map from the canonical cube and the earlier additions used by the witness. A structural analysis of the $n = 10$ certificate isolates a twin-vertex activation lemma and three cube switches. The general clique-extension construction then gives $\text{wsat}(K_n, Q_3) \leq 2n - 3$ for every $n \geq 11$.

Keywords: weak saturation; graph bootstrap percolation; hypercube; extremal graph theory; finite closure systems; computer-assisted proof.

1 Introduction

Let H and G be finite graphs with $H \subseteq G$. A spanning subgraph $F \subseteq G$ is *weakly H -saturated* in G if the edges of $G \setminus F$ can be ordered as $e_1, \dots, e_t$ so that, for every i , the graph

$$F + e_1 + \dots + e_i$$

contains a copy of H in which e_i is an edge. The weak saturation number is

$$\text{wsat}(G, H) = \min\{|E(F)| : F \text{ is weakly } H\text{-saturated in } G\}.$$

For complete hosts, write $\text{wsat}(n, H) = \text{wsat}(K_n, H)$. The parameter originates with Bollobás [1] and has connections with bootstrap percolation, extremal graph theory, and linear-algebraic methods [2, 3, 4].

The problem studied here is

$$\boxed{\text{Determine } \text{wsat}(K_n, Q_3) \text{ for } n \geq 8.}$$

The cube has 8 vertices, 12 edges, and minimum degree 3. The quantity $\text{wsat}(K_n, Q_3)$ is currently listed as an open problem in the Open Problem Garden [9]. The 2025 PCMI project slides of Trakulthongchai, Xu, and Zhu state the bound $\text{wsat}(K_n, Q_3) \leq 2n - 1$ for $n \geq 8$, report the values 15 and 17 at $n = 8$ and $n = 9$, and conjecture equality with $2n - 1$ [8]. The $n = 9$ case is re-evaluated from the closure system defined below. The exact scan of all $\binom{24}{4} = 10,626$ normalized four-edge seeds gives no percolating seed, while a five-edge certificate percolates. An independent Go implementation with explicit edge lists and queue-based activation gives the same four-edge exclusion, and a separate graph-level verifier checks the certificate against all 7,560 labelled copies of Q_3 in K_9 . Consequently, $\text{wsat}(K_9, Q_3) = 16$.

Theorem 1.1 (Main result).

$$\boxed{\text{wsat}(K_9, Q_3) = 16, \quad \text{wsat}(K_{10}, Q_3) = 18, \quad \text{wsat}(K_{11}, Q_3) = 19.}$$

The proof separates the finite generating-set problem from the graph-theoretic bookkeeping. The first-edge normalization supplies a fixed cube core. The closure operator records all legal outside-edge additions. Its finite fixed-point computation is proved correct, and the symmetry reductions are proved complete. The upper bounds are verified edge by edge by explicit cube embeddings.

2 Preliminaries and known bounds

2.1 The cube

Identify $V(Q_3)$ with $\mathbb{F}_2^3$, written as $\{0, 1, \dots, 7\}$ in binary order. Thus

$$xy \in E(Q_3) \iff x \oplus y \text{ has Hamming weight } 1.$$

Consequently

$$|V(Q_3)| = 8, \quad |E(Q_3)| = 12, \quad \delta(Q_3) = 3.$$

Edges are written as concatenated endpoint labels; when the vertex 10 occurs, a thin space separates the labels, so 0 10 denotes the edge $\{0, 10\}$.

The automorphism group has order

$$|\text{Aut}(Q_3)| = 2^3 \cdot 3! = 48,$$

and $\text{Aut}(Q_3)$ is transitive on the twelve cube edges.

Lemma 2.1 (Minimum degree). *Let H be a graph with minimum degree $\delta(H) \geq 2$. If F is weakly H -saturated in K_n , then*

$$\delta(F) \geq \delta(H) - 1.$$

In particular, every weakly Q_3 -saturated spanning graph has minimum degree at least 2.

Proof. Suppose that $v \in V(K_n)$ has $d_F(v) \leq \delta(H) - 2$. Before the first added edge incident with v , the degree of v is still at most $\delta(H) - 2$. If an edge incident with v were added at that stage and completed a copy of H , the image of v in that copy would have at least $\delta(H)$ incident edges, while only the newly added edge can increase the degree of v . Hence such an addition is impossible. Therefore no edge incident with v can ever be added, contradicting saturation. $\square$

2.2 Analytic lower bounds and asymptotic scale

For a graph H with v vertices, ℓ edges, and minimum degree $\delta > 1$, Terekhov and Zhukovskii proved the following lower bound [6]:

$$\text{wsat}(K_n, H) \geq \begin{cases} \frac{\delta}{2}(n - v) + \ell - 1, & H \text{ is 2-edge-connected,} \\ \left\lceil \left(\frac{\delta}{2} - \frac{1}{\delta + 1} \right) (n - v) \right\rceil + \ell - 1, & \text{in general.} \end{cases} \quad (2.1)$$

Since Q_3 is 2-edge-connected,

$$\text{wsat}(K_n, Q_3) \geq \frac{3}{2}(n - 8) + 11. \quad (2.2)$$

For the invariant used in their sharper theorem, set

$$e_i = \min_{\substack{S \subseteq V(Q_3) \\ |S|=i}} (|E(Q_3)| - |E(Q_3 - S)| - 1), \quad 1 \leq i \leq 7. \quad (2.3)$$

A direct calculation gives

$$\begin{array}{c|ccccccc} i & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ \hline e_i & 2 & 4 & 6 & 7 & 9 & 10 & 11 \end{array} \quad (2.4)$$

and hence

$$\min_{1 \leq i \leq 7} \frac{e_i}{i} = \frac{11}{7}. \quad (2.5)$$

The maximal subadditive function in Theorem 2.1 of [6] is obtained from the values e_i by

$$g^*(0) = 0, \quad g^*(t) = \min_{\substack{s \geq 1, i_1, \dots, i_s \in \{1, \dots, 7\} \\ i_1 + \dots + i_s = t}} (e_{i_1} + \dots + e_{i_s}). \quad (2.6)$$

In particular, $g^*(1) = 2$, $g^*(2) = 4$, and $g^*(3) = 6$.

The calculation behind (2.4) follows from the induced-edge maxima in the cube. Let

$$m_i = \max_{|S|=i} |E(Q_3[S])|.$$

Since every vertex of S has degree 3 in Q_3 ,

$$|E(Q_3)| - |E(Q_3 - S)| = 3i - |E(Q_3[S])| = 3i - m_i.$$

Therefore $e_i = 3i - m_i - 1$. The required maxima are

$$\begin{array}{c|ccccccc} i & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ \hline m_i & 0 & 1 & 2 & 4 & 5 & 7 & 9 \end{array}$$

and are attained, for example, by the sets

$$\{0\}, \{0, 1\}, \{0, 1, 2\}, \{0, 1, 2, 3\}, \{0, 1, 2, 3, 4\}, \{0, 1, 2, 3, 4, 5\}, V(Q_3) \setminus \{7\},$$

respectively. Substitution yields the values in (2.4).

Theorem 2.1 of [6], together with the existence of the asymptotic weak saturation slope proved by Alon [5], therefore gives

$$\frac{11}{7} \leq c_{Q_3} := \lim_{n \rightarrow \infty} \frac{\text{wsat}(K_n, Q_3)}{n} \leq 2. \quad (2.7)$$

The upper bound in (2.7) comes from the standard clique-extension construction. The values established here are finite exact values inside this asymptotic interval; (2.2)–(2.6) do not determine $\text{wsat}(K_9, Q_3)$.

3 A finite closure system

3.1 Marked cube and outside edges

Fix the canonical cube

$$C = Q_3[\{0, 1, \dots, 7\}]$$

and the marked edge

$$e^* = 01.$$

For $n \geq 8$ let

$$U_n = E(K_n) \setminus E(C), \quad |U_n| = \binom{n}{2} - 12. \quad (3.1)$$

Every weak saturation order can be normalized so that the first edge is e^* .

Lemma 3.1 (First-edge normalization). *Let F be weakly Q_3 -saturated in K_n , and let e be the first edge in a legal addition order. There is a relabelling of $V(K_n)$ under which*

$$e = e^*, \quad e^* \notin F, \quad E(C) \setminus \{e^*\} \subseteq F,$$

and every edge of F outside $E(C)$ belongs to U_n .

Proof. The first edge e is absent from F and completes a copy Q of Q_3 whose other eleven edges lie in F . Since $\text{Aut}(Q_3)$ is edge-transitive, there is an isomorphism of the marked pair (Q, e) with (C, e^*) . Extend this isomorphism to a permutation of $V(K_n)$. Under the resulting relabelling, $e = e^*$ and $E(C) \setminus \{e^*\} \subseteq F$. Every remaining edge of F lies outside $E(C)$ and hence belongs to U_n . $\square$

Definition 3.2 (Outside closure). *For $S \subseteq U_n$, the closure $\text{cl}_n(S)$ is the least subset $T \subseteq U_n$ satisfying $S \subseteq T$ and the following implication: if $Q \subseteq K_n$ is a copy of Q_3 , $e \in E(Q) \cap U_n$, and*

$$(E(Q) \cap U_n) \setminus \{e\} \subseteq T,$$

then $e \in T$.

For a copy Q of Q_3 , put

$$M(Q) = E(Q) \cap U_n.$$

The copy contributes the implications

$$M(Q) \setminus \{e\} \longrightarrow e, \quad e \in M(Q). \quad (3.2)$$

Copies with the same outside set $M(Q)$ produce the same implications and can be merged.

Lemma 3.3 (Correctness of Algorithm 1). *Algorithm 1 terminates and returns $\text{cl}_n(S)$.*

Proof. Let T_j denote the value of T after j successful insertions. We show by induction that $T_j \subseteq \text{cl}_n(S)$. This holds for $j = 0$ because $S \subseteq \text{cl}_n(S)$. If an edge e is inserted at step $j + 1$, then some rule $A \rightarrow e$ has $A \subseteq T_j$. By the induction hypothesis, $A \subseteq \text{cl}_n(S)$, so the defining closure property gives $e \in \text{cl}_n(S)$. Hence every edge inserted by the algorithm lies in $\text{cl}_n(S)$.

At most $|U_n|$ edges can be inserted, so the algorithm terminates. At termination, if $A \rightarrow e$ is any closure rule with $A \subseteq T$, then the counter for that rule has reached $|A|$ and the target e has been inserted. Thus T is closed under all rules defining $\text{cl}_n(S)$. Since $S \subseteq T$, minimality of $\text{cl}_n(S)$ gives $\text{cl}_n(S) \subseteq T$. Therefore $T = \text{cl}_n(S)$. $\square$

Algorithm 1 Exact closure computation**Require:** seed $S \subseteq U_n$ and the distinct outside masks $\mathcal{M}_n = \{M(Q) : Q \cong Q_3\}$ **Ensure:** $T = \text{cl}_n(S)$

```

1:  $T \leftarrow S$ 
2: insert into  $T$  every target of an implication with empty antecedent, and enqueue the inserted edges
3: enqueue every edge of  $S$ 
4: maintain, for each implication  $A \rightarrow e$ , the number of edges of  $A$  already in  $T$ 
5: while the queue is nonempty do
6:   remove an edge  $f$  from the queue
7:   for all implications  $A \rightarrow e$  with  $f \in A$  do
8:     increment the counter for  $A \rightarrow e$ 
9:     if all edges of  $A$  are in  $T$  and  $e \notin T$  then
10:       add  $e$  to  $T$  and enqueue  $e$ 
11:     end if
12:   end for
13: end while
14: return  $T$ 

```

Lemma 3.4 (Certificate criterion). *Let $F_0 \subseteq K_n$ and let $e_1, \dots, e_t$ be the missing edges of F_0 . Suppose that for each i there is an injective map $\phi_i : V(Q_3) \rightarrow V(K_n)$ satisfying*

$$\phi_i(0)\phi_i(1) = e_i$$

and

$$\phi_i(E(Q_3) \setminus \{01\}) \subseteq E(F_0) \cup \{e_1, \dots, e_{i-1}\}.$$

Then F_0 is weakly Q_3 -saturated in K_n .

Proof. For each i , injectivity of ϕ_i makes its image a copy of Q_3 . The displayed containment shows that every edge of this copy except e_i is present immediately before step i . Hence adding e_i creates a copy of Q_3 . Applying this argument successively proves that the sequence is legal and ends with K_n . $\square$

Lemma 3.5 (Sequential equivalence). *For $S \subseteq U_n$,*

$$(C - e^*) \cup S \text{ is weakly } Q_3\text{-saturated in } K_n \iff \text{cl}_n(S) = U_n.$$

Proof. Suppose that $(C - e^*) \cup S$ is weakly Q_3 -saturated. Let $e_1, \dots, e_t$ be a legal order. By Lemma 3.1, after a relabelling the first edge completes C and is e^* . Moving e^* to the front preserves legality because an edge already available at a given stage remains available after additional edges have been added. Delete this first edge from the order.

Consider an edge $e_i \in U_n$ in the resulting order. Its witnessing cube Q_i has every edge other than e_i present when e_i is added. The edges of Q_i that lie in U_n therefore satisfy one of the implications (3.2). Induction on the order position shows that every outside edge added by the saturation process belongs to $\text{cl}_n(S)$. The final graph is K_n , so $\text{cl}_n(S) = U_n$.

Conversely, suppose that $\text{cl}_n(S) = U_n$. Define rounds by $T_0 = S$ and

$$T_{r+1} = T_r \cup \{e \in U_n : \exists Q \cong Q_3, M(Q) \setminus \{e\} \subseteq T_r\}.$$

The sequence stabilizes because U_n is finite. Add e^* first. For $e \in T_{r+1} \setminus T_r$, its witnessing cube has all other outside edges in T_r , while all cube edges outside U_n lie in $E(C)$ and are present after e^* is added. Thus each edge in a round is legally addable once all earlier rounds have been added. Serializing the rounds gives a legal weak Q_3 -saturation order. $\square$

Lemma 3.6 (Seed identity). *Let*

$$\sigma_n = \min\{|S| : S \subseteq U_n, \text{cl}_n(S) = U_n\}.$$

Then

$$\boxed{\text{wsat}(K_n, Q_3) = 11 + \sigma_n.} \quad (3.3)$$

Proof. Lemma 3.1 puts eleven cube edges in every normalized initial graph. The remaining initial edges form a seed $S \subseteq U_n$. Lemma 3.5 shows that S must percolate, so every weak saturation graph has at least $11 + \sigma_n$ edges. Conversely, a percolating seed together with $C - e^*$ gives a weak saturation graph with $11 + |S|$ initial edges. Minimizing over S proves (3.3). $\square$

Lemma 3.7 (Monotonicity). *If $S \subseteq T \subseteq U_n$, then*

$$\text{cl}_n(S) \subseteq \text{cl}_n(T).$$

Hence every superset of a percolating seed also percolates.

Proof. The defining implications of the closure system are monotone in the current edge set. An induction on closure rounds gives the inclusion. $\square$

3.2 Enumeration of the closure rules

For each n , construct the closure rules by enumerating all copies of Q_3 in K_n . Choose an 8-vertex subset $W \subseteq V(K_n)$ and enumerate the edge sets on W isomorphic to Q_3 . For a copy Q , retain only the edges outside the fixed cube,

$$M(Q) = E(Q) \cap U_n.$$

The copy yields the implications

$$M(Q) \setminus \{e\} \longrightarrow e, \quad e \in M(Q). \quad (3.4)$$

If two cube copies have the same outside mask, their implications coincide, so the two copies can be merged for closure computation.

The number of labelled cube copies is

$$N_n = \binom{n}{8} \frac{8!}{|\text{Aut}(Q_3)|}, \quad (3.5)$$

which gives

$$N_9 = 7,560, \quad N_{10} = 37,800, \quad N_{11} = 138,600. \quad (3.6)$$

The corresponding numbers of distinct outside masks are

$$7,122, \quad 36,506, \quad 135,650. \quad (3.7)$$

Lemma 3.8 (Completeness of the mask system). *Algorithm 2 produces every implication required by Definition 3.2.*

Proof. Every copy $Q \cong Q_3$ in K_n is generated on its 8-vertex set, so its outside mask occurs in $\mathcal{M}_n$. Conversely, every stored mask is the outside mask of a cube copy. The twelve edges of that cube that lie outside U_n belong to $E(C)$, and after e^* is added they are already present. Thus the legality of an outside edge depends on the cube only through $M(Q)$, and merging equal masks preserves the complete implication system. $\square$

Algorithm 2 Construction of the complete outside-rule system**Require:** $n \geq 8$ **Ensure:** distinct outside masks $\mathcal{M}_n$

```

1:  $\mathcal{M}_n \leftarrow \emptyset$ 
2: for all  $W \in \binom{V(K_n)}{8}$  do
3:   for all cube edge sets on  $W$  isomorphic to  $Q_3$  do
4:     compute  $M(Q) = E(Q) \cap U_n$ 
5:     insert  $M(Q)$  into  $\mathcal{M}_n$ 
6:   end for
7: end for
8: return  $\mathcal{M}_n$ 

```

3.3 Symmetry reduction

Define

$$\Gamma_n = \text{Stab}_{\text{Aut}(Q_3)}(e^*) \times S_{\{8, \dots, n-1\}}, \quad (3.4)$$

where the first factor acts on $\{0, \dots, 7\}$ and the second factor permutes the vertices outside the cube. Since an edge orbit of $\text{Aut}(Q_3)$ has size 12,

$$|\text{Stab}_{\text{Aut}(Q_3)}(e^*)| = 48/12 = 4,$$

so

$$|\Gamma_9| = 4, \quad |\Gamma_{10}| = 8, \quad |\Gamma_{11}| = 24. \quad (3.5)$$

Definition 3.9 (Canonical orbit representative). *Fix an ordering of U_n and encode each subset of U_n by its binary incidence mask. Define*

$$\text{can}(S) = \min\{gS : g \in \Gamma_n\}, \quad (3.8)$$

where the minimum is taken in the induced lexicographic order.

Lemma 3.10 (Equivariance). *For every $g \in \Gamma_n$ and $S \subseteq U_n$,*

$$\text{cl}_n(gS) = g \text{cl}_n(S).$$

Proof. If $A \rightarrow e$ is induced by a cube Q , then $gA \rightarrow ge$ is induced by the cube gQ . Hence g preserves the implication system. Induction over the closure rounds gives the stated identity. $\square$

Lemma 3.11 (Canonicalization criterion). *For $S, T \subseteq U_n$,*

$$\text{can}(S) = \text{can}(T) \iff T = gS \text{ for some } g \in \Gamma_n.$$

Proof. If $T = gS$, then S and T have the same orbit and therefore the same canonical representative. Conversely, if the canonical representatives agree, choose $g, h \in \Gamma_n$ with $gS = hT$. Then $T = h^{-1}gS$. $\square$

By Lemmas 3.10 and 3.11, percolation and closure size are constant on each orbit, so one representative from each orbit suffices.

Lemma 3.12 (Layered orbit generation). *Let $\mathcal{R}_k$ contain one representative from every Γ_n -orbit of k -subsets of U_n . The following procedure produces a complete set of representatives for the $(k+1)$ -subsets:*

$$\{\text{can}(R \cup \{e\}) : R \in \mathcal{R}_k, e \in U_n \setminus R\}, \quad (3.6)$$

followed by deduplication.

Proof. Let S be any $(k+1)$ -subset of U_n . Choose $e \in S$ and put $R = S \setminus \{e\}$. By completeness of $\mathcal{R}_k$, some $g \in \Gamma_n$ maps R to its representative $R_0 \in \mathcal{R}_k$. Then $gS = R_0 \cup \{ge\}$ occurs in (3.6), so the orbit of S is represented among the generated candidates. $\square$

Algorithm 3 Complete generation of the next orbit layer**Require:** representatives $\mathcal{R}_k$ of all Γ_n -orbits on k -subsets**Ensure:** representatives $\mathcal{R}_{k+1}$ of all Γ_n -orbits on $(k+1)$ -subsets

```

1:  $\mathcal{C} \leftarrow \emptyset$ 
2: for all  $R \in \mathcal{R}_k$  do
3:   for all  $e \in U_n \setminus R$  do
4:     insert  $\text{can}(R \cup \{e\})$  into  $\mathcal{C}$ 
5:   end for
6: end for
7: deduplicate  $\mathcal{C}$ 
8: return  $\mathcal{C}$ 

```

4 The exact value at $n = 9$

4.1 Upper bound

The seed is

$$S_9 = \{03, 05, 08, 12, 28\}, \quad (4.1)$$

and the initial graph is

$$E(F_9) = E(C) \setminus \{01\} \cup S_9. \quad (4.2)$$

Explicitly,

$$E(F_9) = \{02, 03, 04, 05, 08, 12, 13, 15, 23, 26, 28, 37, 45, 46, 57, 67\}.$$

The twenty missing edges are added in the order

$$\begin{array}{cccccccccccc} 01, & 14, & 48, & 78, & 06, & 16, & 17, & 18, & 24, & 25, \\ 27, & 34, & 35, & 36, & 38, & 47, & 56, & 58, & 68, & 07. \end{array} \quad (4.3)$$

Proposition 4.1. *The graph F_9 is weakly Q_3 -saturated in K_9 . Hence*

$$\text{wsat}(K_9, Q_3) \leq 16.$$

Proof. Apply Lemma 3.4 to the sequence (4.3). For every row of the $n = 9$ witness table in Appendix A, the listed map ϕ_i is injective, sends the canonical edge 01 to the edge added at step i , and maps the other eleven cube edges to initial edges or edges appearing in the dependency column. Hence every row certifies a legal addition.

For step 2, for example,

$$(\phi_2(0), \dots, \phi_2(7)) = (1, 4, 0, 5, 2, 6, 3, 7),$$

and

$$\phi_2(E(Q_3)) = \{14, 01, 12, 45, 46, 05, 03, 57, 26, 23, 67, 37\}.$$

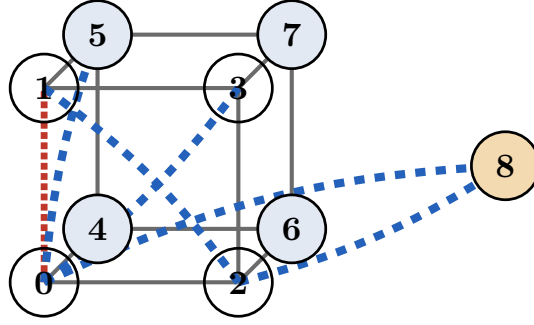
Before step 2, the edge 01 was added at step 1, while the remaining ten non-target edges in this set belong to $E(F_9)$. Thus 14 is legal. The same finite check applies to all twenty rows. The sequence contains every edge outside $E(F_9)$ exactly once, so the terminal graph is K_9 . $\square$

The closure rounds for the seed S_9 have sizes 5, 2, 4, 12, 1:

round	edges activated
0	03, 05, 08, 12, 28
1	14, 48
2	16, 17, 18, 78
3	06, 07, 24, 25, 27, 34, 35, 36, 38, 47, 56, 58
4	68.

(4.4)

F_9 : 16 edges = 11 cube + 5 seed



blue dashed: S_9 ; red dotted: first edge 01; orange: outside vertex

Figure 1: The $n = 9$ upper-bound certificate. Solid edges are the eleven canonical cube edges initially present; the red dotted edge is the marked first edge 01; blue dashed edges form the five-edge seed S_9 ; the orange vertex lies outside the canonical cube.

This round decomposition is obtained directly from the implication system and gives a second serialization of the same certificate.

4.2 Lower bound

Here

$$|U_9| = \binom{9}{2} - 12 = 24.$$

A weak saturation number at most 15 would, by (3.3), give a percolating seed of size at most 4. By Lemma 3.7, it suffices to exclude four-edge seeds, since every seed of size at most 4 is contained in a four-edge seed and percolation is monotone.

Proposition 4.2. *No four-edge subset of U_9 percolates. Consequently,*

$$\sigma_9 \geq 5.$$

Proof. There are exactly $\binom{24}{4} = 10,626$ labelled four-edge subsets of U_9 . The primary C++ enumeration visits each subset exactly once. Burnside’s lemma gives 2,826 orbits under the order-4 group Γ_9 ; the $n = 9$ lower-bound scan does not use this quotient. For each seed, Algorithm 1 computes its exact closure. Every terminal closure has fewer than 24 edges; the largest has 16 edges. Therefore no four-edge seed satisfies $\text{cl}_9(S) = U_9$.

The same four-edge exclusion was recomputed by the independent Go implementation in the supplementary folder. That program constructs the 7,560 labelled cube copies from explicit sorted edge lists, generates the corresponding closure implications, and evaluates the 10,626 seeds with a queue-based activation routine. It returns zero percolating seeds and maximum closure size 16. $\square$

The primary exact scan also evaluates the full five-edge layer. Among the 42,504 five-edge seeds, exactly 6,864 percolate. Together with the explicit five-edge certificate, this gives $\sigma_9 = 5$.

The certificate itself was checked at the graph level against every labelled copy of Q_3 in K_9 . The number of distinct labelled cube edge sets is

$$\frac{\binom{9}{8} 8!}{|\text{Aut}(Q_3)|} = \frac{9 \cdot 40320}{48} = 7,560. \quad (4.5)$$

The initial graph contains none of these 7,560 cubes. Each witness row is verified as one of these labelled cube copies, with its eleven non-target edges present at the required stage, and the final graph is K_9 . This check is implemented separately in `supplement/programs/n9_graph_level_check.py`.

Theorem 4.3.

$$\boxed{\text{wsat}(K_9, Q_3) = 16.}$$

Proof. Proposition 4.1 gives $\text{wsat}(K_9, Q_3) \leq 16$, while Proposition 4.2 and (3.3) give $\sigma_9 \geq 5$ and hence $\text{wsat}(K_9, Q_3) \geq 16$. $\square$

Corollary 4.4. *For every $n \geq 9$,*

$$\text{wsat}(K_n, Q_3) \leq 2n - 2.$$

Hence $\text{wsat}(K_n, Q_3) = 2n - 1$ cannot hold for any $n \geq 9$.

Proof. Apply Proposition 7.1 repeatedly to the 16-edge construction at $n = 9$. Each extension adds two initial edges, giving

$$\text{wsat}(K_n, Q_3) \leq 16 + 2(n - 9) = 2n - 2.$$

$\square$

5 The exact value at $n = 10$

5.1 Upper bound

The seven-edge seed is

$$S_{10} = \{12, 19, 29, 35, 36, 68, 78\}. \quad (5.1)$$

Together with $C - 01$, this gives 18 initial edges. One legal saturation order is

$$\begin{array}{cccccccccc} 01, & 39, & 07, & 17, & 03, & 25, & 47, & 16, & 27, & 56, \\ 24, & 34, & 05, & 06, & 14, & 59, & 49, & 69, & 09, & 79, \\ 28, & 58, & 18, & 38, & 48, & 08, & 89. & & & \end{array} \quad (5.2)$$

The full witness table is in Appendix A.

Proposition 5.1. *The graph defined by (5.1) is weakly Q_3 -saturated in K_{10} . Hence*

$$\text{wsat}(K_{10}, Q_3) \leq 18.$$

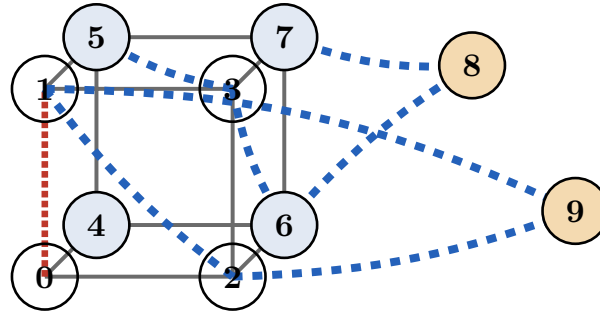
Proof. Lemma 3.4 applies to the order (5.2). For each i , the witness table gives an injective map $\phi_i : V(Q_3) \rightarrow V(K_{10})$ with $\phi_i(0)\phi_i(1) = e_i$ and with every other image of a cube edge belonging to the initial graph or to an earlier step. Hence e_i is legal. The 27 listed additions are exactly the edges missing from the 18-edge initial graph, so the terminal graph is K_{10} . $\square$

5.2 A structural activation lemma

The early part of the certificate has a simple interpretation.

Lemma 5.2 (Twin activation). *Let $Q \cong Q_3$ be a copy in a current graph, let $v \in V(Q)$ have neighbours a, b, c in Q , and let $x \notin V(Q)$. If xa and xb are already present, then xc is legal.*

F_{10} : 18 edges = 11 cube + 7 seed



blue dashed: S_{10} ; red dotted: first edge 01; orange: outside vertices

Figure 2: The $n = 10$ upper-bound certificate. The canonical cube is on vertices 0–7; vertices 8 and 9 are outside the cube.

Proof. The graph obtained from Q by deleting v and inserting x , with x adjacent to a, b, c , is isomorphic to Q_3 : it is the cube obtained by replacing the vertex v by a vertex with the same three neighbours. The nine edges of $Q - v$ remain present and the three edges xa, xb, xc form the three edges incident with x . Thus adding xc completes a cube. $\square$

In the $n = 10$ seed, 19 and 29 are present initially. Applying Lemma 5.2 with $v = 0$, whose cube neighbours are 1, 2, 4, gives 49. Applying it with $v = 3$, whose cube neighbours are 1, 2, 7, gives 79. The displayed certificate also creates 39 immediately after 01; this edge is not required for the activations 49, 79, and 07.

Once 01, 19, 29, and 49 are present, the cube with vertex map

$$(0, 1, 2, 3, 4, 5, 6, 7) \mapsto (0, 7, 1, 3, 4, 6, 9, 2)$$

has edge set

$$\{01, 04, 07, 13, 19, 23, 26, 29, 37, 46, 49, 67\}.$$

All edges other than 07 are then present, so 07 is legal. Thus the first structural activation can be ordered as

$$01, 39, 49, 79, 07.$$

The edge 07 has the following three relevant cube switches:

$$\begin{aligned} C_1 &= C - \{01, 67\} + \{07, 16\}, \\ C_2 &= C - \{02, 57\} + \{07, 25\}, \\ C_3 &= C - \{04, 37\} + \{07, 34\}. \end{aligned} \tag{5.3}$$

Each C_i is a copy of Q_3 . Explicit cube maps from the canonical cube are, respectively,

$$(0, 1, 2, 3, 4, 5, 6, 7) \mapsto (1, 6, 3, 2, 5, 4, 7, 0),$$

$$(0, 1, 2, 3, 4, 5, 6, 7) \mapsto (2, 5, 3, 1, 6, 4, 7, 0),$$

and

$$(0, 1, 2, 3, 4, 5, 6, 7) \mapsto (3, 4, 1, 5, 2, 6, 0, 7).$$

The first map replaces the cube edges 01, 67 by 07, 16; the second replaces 02, 57 by 07, 25; and the third replaces 04, 37 by 07, 34. Once 07 is present, each switch has exactly one absent edge, so 16, 25, and 34 are legal in succession. The remaining edges on the eight cube vertices then follow from the certificate rounds.

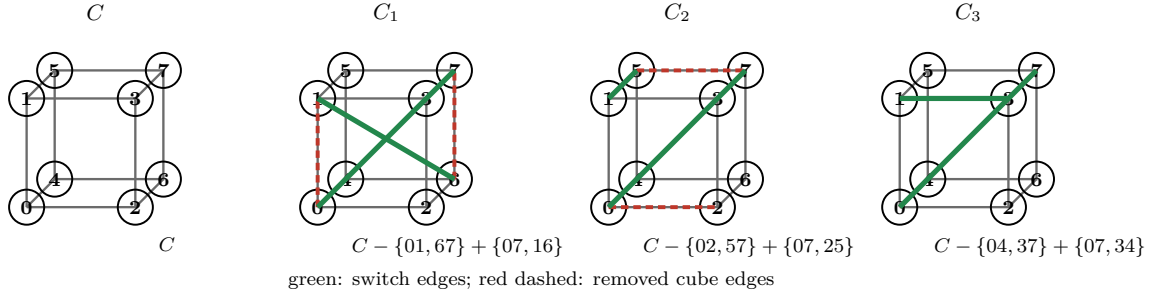


Figure 3: The three cube switches used after 07 appears. In each panel the green edge is the added edge; red dashed edges are the two cube edges replaced by the switch.

5.3 Closure layers

The closure of S_{10} has three closure layers beyond the seed:

round	edges activated
0	12, 19, 29, 35, 36, 68, 78
1	39, 49, 79
2	03, 07, 14, 16, 17, 24, 25, 27, 34
3	05, 06, 08, 09, 18, 28, 38, 47, 48, 56, 58, 59, 69, 89.

(5.4)

An edge in round r depends only on edges in rounds at most $r - 1$, so any ordering within one round is legal. This gives a direct round-by-round verification of the certificate closure.

5.4 Lower bound

Here

$$|U_{10}| = \binom{10}{2} - 12 = 33.$$

To prove $\text{wsat}(K_{10}, Q_3) \geq 18$, it suffices to exclude six-edge seeds: by Lemma 3.7, every percolating seed of size at most 6 would extend to a percolating six-edge seed.

There are

$$\binom{33}{6} = 1,107,568$$

six-edge seeds. The enumeration canonicalizes each seed under the order-8 group Γ_{10} . By Lemma 3.10, two seeds in the same orbit have identical closure size, so one representative per orbit suffices. The resulting complete set has exactly

$$146, 131$$

representatives. None percolates; the maximum closure contains 25 of the 33 outside edges.

Proposition 5.3. *No six-edge subset of U_{10} percolates. Hence $\sigma_{10} \geq 7$.*

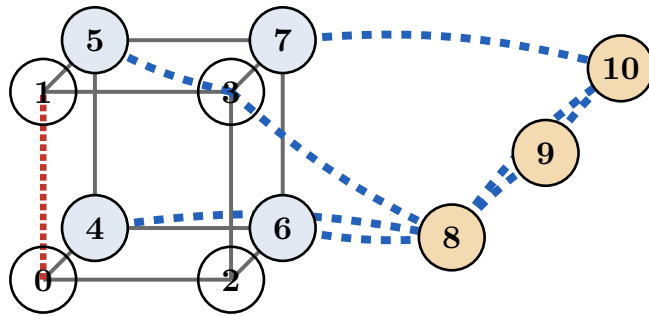
Proof. The enumeration contains one representative from every Γ_{10} -orbit of six-edge subsets. Lemma 3.10 preserves percolation across each orbit. The exact closure routine returns fewer than 33 outside edges for every representative. Therefore no six-edge seed percolates. $\square$

Theorem 5.4.

$\text{wsat}(K_{10}, Q_3) = 18.$

Proof. The proposition gives $\sigma_{10} \geq 7$ and hence $\text{wsat}(K_{10}, Q_3) \geq 18$. Proposition 5.1 gives the reverse inequality. $\square$

F_{11} : 19 edges = 11 cube + 8 seed



blue dashed: S_{11} ; red dotted: first edge 01; orange: outside vertices

Figure 4: The $n = 11$ upper-bound certificate. The orange vertices 8, 9, 10 lie outside the canonical cube.

6 The exact value at $n = 11$

6.1 Upper bound

The eight-edge seed is

$$S_{11} = \{35, 38, 48, 68, 710, 89, 310, 910\}. \quad (6.1)$$

Together with $C - 01$, it gives 19 initial edges. A legal addition order is

$$\begin{array}{cccccccccc} 01, & 08, & 25, & 07, & 09, & 16, & 18, & 03, & 05, & 06, \\ 010, & 12, & 14, & 17, & 19, & 110, & 24, & 27, & 28, & 29, \\ 210, & 34, & 36, & 39, & 47, & 49, & 410, & 56, & 58, & 59, \\ 510, & 69, & 610, & 78, & 79, & 810. \end{array} \quad (6.2)$$

Proposition 6.1. *The graph defined by (6.1) is weakly Q_3 -saturated in K_{11} . Hence*

$$\text{wsat}(K_{11}, Q_3) \leq 19.$$

Proof. Lemma 3.4 applies to the order (6.2). For every listed edge e_i , the witness table gives an injective map $\phi_i : V(Q_3) \rightarrow V(K_{11})$ taking 01 to e_i and taking every other cube edge to an initially present edge or an edge e_j with $j < i$. Hence each addition creates a copy of Q_3 . The 36 additions exhaust the edges missing from the 19-edge initial graph, so the terminal graph is K_{11} . $\square$

6.2 Lower bound and orbit generation

For $n = 11$,

$$|U_{11}| = \binom{11}{2} - 12 = 43.$$

A bound of 18 would require a percolating seed of size at most 7. By Lemma 3.7, any percolating seed of size at most 7 would extend to a percolating seven-edge seed. It therefore suffices to exclude seven-edge seeds. The raw number of seven-edge seeds is

$$\binom{43}{7} = 32,224,114.$$

The group Γ_{11} has order 24. The six-edge orbit layer contains 300,563 representatives. Applying Lemma 3.12 to that layer produces 11,120,831 seven-edge candidates before canonical deduplication and exactly

$$1,498,208$$

representatives afterwards.

Proposition 6.2. *No seven-edge subset of U_{11} percolates. Hence $\sigma_{11} \geq 8$.*

Proof. Start with the complete set of Γ_{11} -orbit representatives at size 6. Lemma 3.12 shows that adjoining every possible new edge and canonicalizing yields a complete set of orbit representatives at size 7. There are 11,120,831 generated children before deduplication and 1,498,208 representatives afterwards. By Lemma 3.10, closure size is constant on each orbit. Algorithm 1 gives no full closure among these representatives; the largest terminal closure has 33 of the 43 outside edges. Hence no seven-edge seed percolates. $\square$

6.3 Burnside verification of the orbit counts

For an element $g \in \Gamma_{11}$ acting on U_{11} with cycle type

$$1^{a_1} 2^{a_2} 3^{a_3} 6^{a_6},$$

the number of fixed k -subsets is

$$\text{Fix}_k(g) = [x^k](1+x)^{a_1}(1+x^2)^{a_2}(1+x^3)^{a_3}(1+x^6)^{a_6}. \quad (6.3)$$

The full cycle-type calculation is Appendix B. Its aggregate values are

$$\begin{aligned} \sum_{g \in \Gamma_{11}} \text{Fix}_6(g) &= 7,213,512, \\ \sum_{g \in \Gamma_{11}} \text{Fix}_7(g) &= 35,956,992. \end{aligned}$$

Burnside's lemma therefore gives

$$|X_6/\Gamma_{11}| = \frac{7,213,512}{24} = 300,563, \quad (6.4)$$

and

$$|X_7/\Gamma_{11}| = \frac{35,956,992}{24} = 1,498,208. \quad (6.5)$$

These values agree with the orbit-generation counts used in the lower-bound search.

Theorem 6.3.

$$\boxed{\text{wsat}(K_{11}, Q_3) = 19.}$$

Proof. The lower-bound proposition gives $\sigma_{11} \geq 8$, hence $\text{wsat}(K_{11}, Q_3) \geq 19$. Proposition 6.1 gives the reverse inequality. $\square$

7 A general extension bound

The following construction is standard and appears in this form in Terekhov and Zhukovskii [6].

Proposition 7.1 (Clique extension). *For every $n \geq 8$,*

$$\text{wsat}(K_{n+1}, Q_3) \leq \text{wsat}(K_n, Q_3) + 2. \quad (7.1)$$

Proof. Let F be weakly Q_3 -saturated in K_n , and let x be a new vertex. Choose distinct $a, b \in V(F)$ and include xa and xb initially. First follow a legal order taking F to K_n on the old vertex set. Fix $y \in V(F) \setminus \{a, b\}$. Since $n \geq 8$, four further old vertices can be chosen distinct from a, b, y . In the completed old clique choose a copy $Q \cong Q_3$ containing a vertex v whose neighbours are a, b, y and whose remaining four vertices are chosen arbitrarily from those four vertices. Replacing v by x gives a cube whose three edges incident with x are xa, xb, xy . The first two are present and the other nine cube edges lie in the completed old clique. Lemma 5.2 therefore makes xy legal. Repeating for every $y \in V(F) \setminus \{a, b\}$ completes all edges incident with x . Exactly two edges incident with x were initially present, so (7.1) follows. This is the Q_3 specialization of the extension construction in Terekhov and Zhukovskii [6]. $\square$

Corollary 7.2. *From $\text{wsat}(K_{11}, Q_3) = 19$,*

$$\boxed{\text{wsat}(K_n, Q_3) \leq 2n - 3 \quad (n \geq 11).} \quad (7.2)$$

Proof. Apply Proposition 7.1 repeatedly. Each added vertex costs two initial edges. $\square$

The sequence of exact values obtained here is therefore

$$\begin{array}{c|cccc} n & 8 & 9 & 10 & 11 \\ \hline \text{wsat}(K_n, Q_3) & 15 & 16 & 18 & 19 \end{array} \quad (7.3)$$

where the $n = 8$ value is reproduced by the independent sanity check in the supplementary folder. The increments from $n = 8$ to 11 are 1, 2, 1.

8 Analytic perspective

The exact lower bound at $n = 9$ is a finite exhaustive statement. The invariant in (2.3) also gives finite lower bounds through Theorem 2.1 of [6]. The values $g^*(1), g^*(2), g^*(3)$ are given explicitly in (2.6), so that

$$\text{wsat}(K_9, Q_3) \geq 13, \quad \text{wsat}(K_{10}, Q_3) \geq 15, \quad \text{wsat}(K_{11}, Q_3) \geq 17.$$

The asymptotic coefficient obtained from the same invariant is $11/7$ in (2.7); the finite computations here improve these analytic bounds by 3, 3, and 2, respectively.

A direct linear-algebraic proof of $\sigma_9 \geq 5$ would require a vector assignment to the 24 outside edges for which every cube implication is represented by a linear dependence. In a formulation adapted to the closure system, one seeks vectors w_e such that, whenever M is a nonempty outside-edge mask of a cube and $e \in M$,

$$w_e \in \text{span}\{w_f : f \in M \setminus \{e\}\}.$$

A span of dimension at least 5 would imply that every percolating seed has at least five edges. No such representation is established here. The limitations of the direct weak-saturation rank method for general graphs are established in [7].

The combinatorial lower-bound framework of Terekhov and Zhukovskii supplies the analytic slope bound in (2.7). Obtaining a direct finite inequality at $n = 9$ that reaches 16 remains a natural problem for further work.

9 Verification architecture and reproducibility

The finite lower-bound computations use the complete implication system of Section 3.2. Each edge of U_n is assigned an index. In the bit-mask implementation, a seed or closure is an integer

Table 1: Finite lower-bound searches. The $n = 9$ scan evaluates all labelled seeds; the orbit count is included for comparison.

n	$ U_n $	seed size	seeds evaluated	orbit representatives	percolating
9	24	4	10,626	2,826	0
10	33	6	1,107,568	146,131	0
11	43	7	32,224,114	1,498,208	0

Table 2: Graph-level checks of the three saturation certificates.

n	Q_3 copies checked	initial copies	certificate additions
9	7,560	0	20
10	37,800	0	27
11	138,600	0	36

with one bit per outside edge. A rule $A \rightarrow e$ is stored as an antecedent bit mask, a target index, and an incidence list for the edges of A . When an edge enters the closure, the counters of the rules containing that edge are incremented. A rule is fired exactly when its counter reaches $|A|$. Because each edge enters at most once, the queue performs a finite number of successful insertions, and Lemma 3.3 identifies the resulting set with the mathematical closure.

Canonicalization evaluates the action of every element of the finite group Γ_n on the incidence mask and selects the least resulting mask. By Lemma 3.11, equality of canonical masks is equivalent to membership in the same orbit. Algorithm 3 therefore generates every orbit at the next seed size.

The independent $n = 9$ Go implementation stores endpoint pairs explicitly and represents rule antecedents as sorted edge lists. It uses a queue of newly active edges and recomputes the same closure relation without the bit-mask state representation of the primary C++ verifier.

Certificate discovery uses a fixed-cardinality local search on subsets of U_n . A move exchanges one seed edge with one edge outside the seed and recomputes the exact closure; the score is the closure size. The implementation also uses simulated annealing with random restarts. Each candidate seed is subjected to the explicit witness-map verifier before it is used as an upper-bound certificate. The heuristic search is not used in the lower-bound arguments.

For $n = 9$, the primary lower-bound verifier is written in C++ and uses integer bit masks for the 24 outside edges. The independent lower-bound verifier is written in Go and uses explicit sorted edge lists and adjacency-style rule indexing. The graph-level certificate checker is written in Python and enumerates the 7,560 labelled cubes as edge sets. Thus the $n = 9$ lower bound and the $n = 9$ upper-bound certificate are checked by implementations with different data structures and independent code paths.

For $n = 10$, the exact orbit search tests all 146,131 representatives of six-edge subsets under Γ_{10} . A separately implemented recheck reconstructs the orbit layer and closure relation independently and returns the same representative count and no percolating seed. Separate Python and C++ programs verify the upper-bound certificate. For $n = 11$, the seven-edge orbit layer is generated from the six-edge layer, canonicalized under Γ_{11} , and then tested by the C++ closure routine. The Burnside calculation is implemented independently in Python.

The corresponding maximum closure sizes are 16 for the $n = 9$ four-edge search, 25 for the $n = 10$ six-edge search, and 33 for the $n = 11$ seven-edge search.

The constructive certificates are independently checked at the graph level by enumerating all labelled copies of Q_3 in the corresponding complete host.

The labelled-copy count is the number of explicit cube edge sets checked by the graph-level verifier. The distinct-mask count is the number of unique implication masks retained by

Table 3: Labelled cube copies and distinct outside masks used by the closure system.

n	labelled Q_3 copies	distinct outside masks
9	7,560	7,122
10	37,800	36,506
11	138,600	135,650

Algorithm 2; equal masks are merged because they induce identical closure rules.

All machine-readable certificates, source files, exact execution logs, and hashes are included under `supplement/`. The root-level `README.md` gives the commands for rebuilding the paper and rerunning every theorem-bearing computation. The folder separates manuscript source, theorem-bearing programs, certificates, execution records, and file hashes so that every numerical claim has a corresponding reproducibility record.

10 Relation to prior work and open problems

The standard clique-extension upper bound gives slope at most 2 for Q_3 , while the Terekhov–Zhukovskii invariant gives the asymptotic lower bound $11/7$. The finite values

$$15, 16, 18, 19$$

therefore leave a substantial gap between the exact small cases and the known asymptotic lower bound.

The correction at $n = 9$ changes the finite picture immediately: the conjectural value $2n - 1$ is 17 at $n = 9$, whereas the certificate and exhaustive lower-bound proof give 16. The same 16-edge certificate extends to an 18-edge upper bound at $n = 10$, and the 19-edge $n = 11$ certificate improves on the resulting 20-edge bound.

Several questions remain.

1. Determine $\text{wsat}(K_n, Q_3)$ for the next values of n , beginning with $n = 12$.
2. Determine the asymptotic slope c_{Q_3} in (2.7), currently bounded by $11/7 \leq c_{Q_3} \leq 2$.
3. Find structural families that explain the transition from the exceptional finite values to a general upper bound below $2n - 3$, or prove that the $2n - 3$ family is asymptotically optimal.
4. Develop a sharp analytic lower bound specialized to Q_3 . In particular, a linear-algebraic or other invariant attaining 16 at $n = 9$ would replace the finite exclusion of all 10, 626 four-edge seeds by a direct inequality.

11 Conclusion

The exact values are

$$\boxed{\text{wsat}(K_9, Q_3) = 16, \quad \text{wsat}(K_{10}, Q_3) = 18, \quad \text{wsat}(K_{11}, Q_3) = 19.}$$

The upper bounds are given by the explicit witness tables, and the lower bounds by the complete finite closure searches. The $n = 11$ construction gives $\text{wsat}(K_n, Q_3) \leq 2n - 3$ for every $n \geq 11$.

Code and data availability. Source code, machine-readable certificates, orbit representatives, and verification records for all computations reported in this paper are available in the public repository [10]. The release `v1.0.0-paper` is archived at Zenodo under the DOI given in [10].

References

- [1] B. Bollobás, Weakly k -saturated graphs, in *Beiträge zur Graphentheorie*, B. G. Teubner Verlagsgesellschaft, Leipzig, 1968, pp. 25–31.
- [2] J. Balogh, B. Bollobás, R. Morris, and O. Riordan, Linear algebra and bootstrap percolation, *Journal of Combinatorial Theory, Series A* 119 (2012), 1328–1335, doi:10.1016/j.jcta.2012.03.005.
- [3] N. Morrison, J. A. Noel, and A. Scott, Saturation in the hypercube and bootstrap percolation, *Combinatorics, Probability and Computing* 26 (2017), 78–98, <https://doi.org/10.1017/S0963548316000122>.
- [4] G. Kronenberg, T. Martins, and N. Morrison, Weak saturation numbers of complete bipartite graphs in the clique, *Journal of Combinatorial Theory, Series A* 178 (2021), 105357, doi:10.1016/j.jcta.2020.105357.
- [5] N. Alon, An extremal problem for sets with applications to graph theory, *Journal of Combinatorial Theory, Series A* 40 (1985), 82–89, doi:10.1016/0097-3165(85)90048-2.
- [6] N. Terekhov and M. Zhukovskii, Weak saturation in graphs: a combinatorial approach, *Journal of Combinatorial Theory, Series B* 172 (2025), 146–167, doi:10.1016/j.jctb.2024.12.007.
- [7] N. Terekhov and M. Zhukovskii, Weak saturation rank: a failure of the linear algebraic approach to weak saturation, *Combinatorica* 45 (2025), doi:10.1007/s00493-025-00174-y.
- [8] T. Trakulthongchai, D. Xu, and K. Zhu, *Weak saturation number of the 3-cube in the complete graph*, Park City Mathematics Institute, Summer 2025 project slides, https://www.ias.edu/sites/default/files/Trakulthongchai_Xu_Zhu.pdf.
- [9] Open Problem Garden, *Weak saturation of the cube in the clique*, problem OPG-60013, https://www.openproblemgarden.org/op/weak_saturation_of_the_cube_in_the_clique.
- [10] V. Saha, *Computational materials for Weak Saturation of the 3-Cube in Complete Graphs: Exact Values at Orders 9,10,11*, version 1.0.0-paper, Zenodo, 2026, doi:<https://doi.org/10.5281/zenodo.23121234>.

A Complete certificate witness tables

For each certificate, row i records the added edge e_i , an injective map

$$\phi_i = (\phi_i(0), \dots, \phi_i(7)),$$

and the earlier certificate steps used by the witness. The canonical cube edge set is

$$E(Q_3) = \{01, 02, 04, 13, 15, 23, 26, 37, 45, 46, 57, 67\}.$$

Thus a row is checked by mapping this edge set through ϕ_i . The target edge is $\phi_i(0)\phi_i(1)$; the dependency column identifies precisely those witness edges created during the certificate. Every other witness edge is initial.

$n = 9$

step	edge	$\phi(0), \dots, \phi(7)$	earlier steps used
1	01	(0, 1, 2, 3, 4, 5, 6, 7)	-
2	14	(1, 4, 0, 5, 2, 6, 3, 7)	1
3	48	(4, 8, 5, 0, 6, 2, 7, 3)	-
4	78	(7, 8, 3, 2, 5, 4, 0, 1)	1, 2, 3
5	06	(0, 6, 3, 2, 5, 4, 7, 8)	3, 4
6	16	(1, 6, 3, 2, 5, 4, 0, 8)	3
7	17	(1, 7, 2, 3, 4, 5, 6, 0)	2, 5
8	18	(1, 8, 3, 2, 5, 4, 0, 6)	3, 5
9	24	(2, 4, 3, 1, 6, 0, 7, 5)	2, 5
10	25	(2, 5, 3, 1, 6, 4, 0, 8)	3, 5, 8
11	27	(2, 7, 1, 3, 4, 5, 6, 0)	5, 6, 9
12	34	(3, 4, 1, 6, 2, 5, 8, 0)	5, 6, 8, 10
13	35	(3, 5, 4, 0, 7, 1, 2, 6)	5, 6, 7, 9, 11, 12
14	36	(3, 6, 4, 1, 5, 0, 2, 8)	2, 5, 6, 8, 9, 10, 12, 13
15	38	(3, 8, 4, 1, 5, 0, 2, 6)	2, 5, 6, 8, 9, 10, 12, 13
16	47	(4, 7, 3, 2, 5, 1, 0, 6)	5, 6, 7, 11, 12
17	56	(5, 6, 3, 1, 4, 0, 2, 8)	5, 6, 8, 9, 13
18	58	(5, 8, 3, 1, 4, 0, 2, 6)	5, 6, 8, 9, 13
19	68	(6, 8, 2, 1, 3, 0, 4, 5)	8, 9, 12, 14
20	07	(0, 7, 5, 2, 6, 1, 3, 4)	2, 5, 6, 7, 9, 10, 11, 12, 13, 14

$n = 10$

step	edge	$\phi(0), \dots, \phi(7)$	earlier steps used
1	01	(0, 1, 2, 3, 4, 5, 6, 7)	-
2	39	(3, 9, 5, 1, 6, 2, 4, 0)	1
3	07	(0, 7, 1, 5, 2, 6, 9, 3)	1, 2
4	17	(1, 7, 2, 0, 3, 5, 6, 4)	3
5	03	(0, 3, 2, 1, 4, 5, 6, 7)	4
6	25	(2, 5, 6, 4, 9, 1, 3, 0)	1, 2, 5
7	47	(4, 7, 5, 1, 6, 3, 2, 0)	1, 4, 5, 6
8	16	(1, 6, 0, 2, 7, 4, 3, 5)	1, 4, 5, 6, 7
9	27	(2, 7, 0, 3, 5, 4, 1, 6)	1, 5, 6, 7, 8
10	56	(5, 6, 1, 3, 4, 7, 0, 2)	1, 7, 9
11	24	(2, 4, 0, 7, 3, 6, 1, 5)	1, 3, 7, 10
12	34	(3, 4, 0, 2, 6, 7, 1, 5)	1, 5, 6, 7, 8, 11
13	05	(0, 5, 1, 6, 2, 4, 3, 7)	1, 7, 8, 10, 11
14	06	(0, 6, 1, 5, 2, 4, 3, 7)	1, 7, 10, 11
15	14	(1, 4, 0, 2, 5, 7, 3, 6)	1, 5, 7, 11
16	59	(5, 9, 4, 2, 6, 3, 1, 0)	1, 2, 5, 8, 10, 11, 15
17	49	(4, 9, 5, 2, 6, 3, 1, 0)	1, 2, 5, 6, 8
18	69	(6, 9, 2, 4, 3, 5, 0, 1)	1, 5, 11, 15, 16, 17
19	09	(0, 9, 1, 4, 2, 5, 3, 6)	1, 6, 10, 15, 16, 17
20	79	(7, 9, 2, 4, 3, 5, 0, 1)	1, 5, 9, 11, 15, 16, 17
21	28	(2, 8, 0, 6, 3, 7, 1, 4)	1, 7, 14, 15

continued on next page

step	edge	$\phi(0), \dots, \phi(7)$	earlier steps used
22	58	(5, 8, 1, 6, 2, 7, 0, 3)	1, 5, 6, 8, 9
23	18	(1, 8, 0, 2, 4, 6, 3, 5)	1, 5, 6, 10, 12, 15, 21
24	38	(3, 8, 0, 2, 5, 6, 1, 4)	1, 5, 10, 11, 15, 21
25	48	(4, 8, 1, 5, 2, 6, 0, 3)	1, 5, 11, 15, 22
26	08	(0, 8, 1, 4, 2, 5, 3, 6)	1, 6, 10, 15, 22, 25
27	89	(8, 9, 2, 4, 3, 5, 0, 1)	1, 5, 11, 15, 16, 17, 21, 24

$n = 11$

step	edge	$\phi(0), \dots, \phi(7)$	earlier steps used
1	01	(0, 1, 2, 3, 4, 5, 6, 7)	-
2	08	(0, 8, 1, 3, 4, 6, 5, 7)	1
3	25	(2, 5, 0, 1, 6, 7, 8, 3)	1, 2
4	07	(0, 7, 1, 3, 4, 6, 5, 2)	1, 3
5	09	(0, 9, 4, 8, 7, 10, 5, 3)	4
6	16	(1, 6, 0, 4, 3, 7, 2, 5)	1, 3
7	18	(1, 8, 0, 9, 5, 3, 7, 10)	1, 4, 5
8	03	(0, 3, 1, 8, 2, 5, 6, 4)	1, 3, 6, 7
9	05	(0, 5, 3, 2, 8, 4, 1, 6)	2, 3, 6, 7, 8
10	06	(0, 6, 3, 2, 8, 4, 1, 5)	2, 3, 7, 8
11	010	(0, 10, 6, 7, 8, 3, 1, 5)	2, 6, 7, 10
12	12	(1, 2, 0, 5, 8, 6, 3, 7)	1, 3, 7, 8, 9
13	14	(1, 4, 0, 5, 6, 8, 2, 3)	1, 6, 9
14	17	(1, 7, 0, 3, 4, 6, 5, 2)	1, 3, 8, 9, 13
15	19	(1, 9, 0, 10, 6, 8, 2, 3)	1, 6, 11
16	110	(1, 10, 0, 3, 6, 7, 2, 5)	1, 3, 6, 8
17	24	(2, 4, 0, 5, 6, 8, 1, 3)	1, 6, 9
18	27	(2, 7, 0, 3, 4, 6, 1, 8)	1, 7, 8, 13, 17
19	28	(2, 8, 0, 3, 4, 6, 1, 7)	1, 8, 13, 14, 17
20	29	(2, 9, 0, 1, 4, 8, 5, 3)	1, 9, 15, 17
21	210	(2, 10, 0, 3, 6, 7, 1, 5)	1, 6, 8
22	34	(3, 4, 0, 1, 8, 6, 2, 7)	1, 8, 13, 14, 18, 19
23	36	(3, 6, 0, 2, 4, 8, 1, 9)	1, 8, 13, 15, 20, 22
24	39	(3, 9, 0, 2, 6, 8, 1, 4)	1, 6, 8, 13, 17, 20, 23
25	47	(4, 7, 1, 5, 2, 6, 0, 3)	1, 8, 13, 17, 23
26	49	(4, 9, 1, 8, 2, 10, 0, 3)	1, 7, 8, 13, 17, 21
27	410	(4, 10, 1, 7, 2, 9, 0, 3)	1, 8, 13, 14, 17, 20, 24
28	56	(5, 6, 1, 4, 3, 7, 0, 2)	1, 8, 13, 17, 18
29	58	(5, 8, 1, 4, 3, 6, 0, 2)	1, 8, 13, 17, 23
30	59	(5, 9, 1, 4, 3, 8, 0, 2)	1, 8, 13, 17, 19, 26
31	510	(5, 10, 1, 4, 3, 7, 0, 2)	1, 8, 13, 17, 18, 27
32	69	(6, 9, 2, 4, 3, 5, 0, 1)	1, 8, 13, 17, 23, 26, 30
33	610	(6, 10, 2, 4, 3, 5, 0, 1)	1, 8, 13, 17, 23, 27, 31
34	78	(7, 8, 2, 4, 3, 5, 0, 1)	1, 8, 13, 17, 18, 29
35	79	(7, 9, 2, 4, 3, 5, 0, 1)	1, 8, 13, 17, 18, 26, 30
36	810	(8, 10, 2, 4, 3, 5, 0, 1)	1, 8, 13, 17, 19, 27, 31

B Burnside table

For $g \in \Gamma_{11}$, let $a_j(g)$ denote the number of j -cycles in the permutation induced by g on the 43 outside edges. The fixed k -subsets are counted by

$$\text{Fix}_k(g) = [x^k](1+x)^{a_1(g)}(1+x^2)^{a_2(g)}(1+x^3)^{a_3(g)}(1+x^6)^{a_6(g)}. \quad (\text{B.1})$$

The complete calculation is

cycle type on U_{11}	multiplicity	Fix ₆	Fix ₇
1^{43}	1	6,096,454	32,224,114
$1^{23}2^{10}$	3	301,834	1,043,770
$1^{11}2^{16}$	3	12,902	33,682
1^22^{21}	3	1,330	1,330
$1^{16}3^9$	2	13,084	28,396
$1^62^53^56^2$	2	498	862
$1^22^93^64$	2	74	106
2^83^64	2	60	28
1^32^{20}	4	1,710	3,610
$1^{21}2^{11}$	1	131,814	416,734
1^52^{19}	1	2,774	6,574
total	24	7,213,512	35,956,992

(B.2)

Consequently, Burnside's lemma gives

$$|X_6/\Gamma_{11}| = \frac{7,213,512}{24} = 300,563, \quad |X_7/\Gamma_{11}| = \frac{35,956,992}{24} = 1,498,208. \quad (\text{B.3})$$

The first total equals $\binom{43}{6}$ after weighting by stabilizers, while the second total similarly accounts for the 7-subset action. These orbit counts agree with the representatives produced by Algorithm 3.

C Finite-search data and verification records

The supplementary material contains the executable verification programs, machine-readable certificates, orbit representatives, and execution records used for the finite computations.

- **Certificates:** `supplement/certificates/` contains the complete machine-readable certificates and witness tables.
- **$n = 9$ verification:** `supplement/programs/n9_exact_verifier.cpp`, `n9_independent_lower.go`, and `n9_graph_level_check.py` provide distinct verification layers.
- **Lower bounds for $n = 10, 11$:** `supplement/programs/n10_lower_orbit_recheck.cpp` and `n11_lower_orbit7.cpp`.
- **Burnside verification:** `supplement/programs/n11_burnside.py`.
- **Certificate verification:** Python and C++ verifiers in `supplement/programs/`.
- **Records and hashes:** `logs/` contains execution outputs and `SHA256SUMS.txt` contains file hashes.

The finite verification specification in `PROOF_SPEC.md` records the edge sets, closure rules, orbit actions, certificate conditions, and finite statements used by the verification programs.